

Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

- A. follicle-stimulating hormone
- B. oxytocin
- C. pituitary gland
- D. hypothalamus
- E. gonadotropin-releasing hormone
- F. luteinizing hormone
- G. testosterone

Reasoning + Answer:

<think>

Given the components of the male reproductive system and their functions , it's clear that the feedback loop involves regulatory mechanisms that maintain hormone levels and optimize reproductive function .

- The hypothalamus plays a crucial role in the production of gonadotropin-releasing hormone (GnRH), which stimulates the pituitary gland .
- The pituitary gland releases follicle-stimulating hormone (FSH) and luteinizing hormone (LH), which in turn stimulate the gonads (testes in males).
- Oxytocin is associated with social bonding , childbirth , and uterine contractions , rather than the direct regulation of sex hormone production .
- Testosterone is a sex hormone , but it's not a part of the feedback loop regulating its production .
- The gonadotropin-releasing hormone (GnRH) from the hypothalamus stimulates the release of LH and FSH from the pituitary gland , which in turn regulates the production of testosterone and other sex hormones in the gonads .
- LH plays a key role in the feedback loop by stimulating the Leydig cells in the testes to produce testosterone .
- Thus , the component that is not part of the feedback loop is oxytocin .

</think>

<answer> B.

